Actuator Operation Manual

Servo Stepper

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Warranty Statement

Under normal use, if the product malfunctions or is damaged, Changingtek Technology provides warranty service within the warranty period (please refer to the order form for product warranty period details). After the warranty period, repair fees will apply. Within the warranty period, repair fees will apply for product damage caused by the following situations.

- Damage to the product caused by failure to operate the product according to the instructions in the manual.
- Product damage caused by fire, flood, or abnormal voltage.
- Damage to the product caused by using it for non-normal functions.
- Damage to the product caused by exceeding the product's specified usage range.
- Secondary damage to products caused by force majeure (natural disasters, earthquakes, lightning strikes).

Service fees are calculated according to the manufacturer's unified standards. If there is a contract, the contract shall take precedence.

Version change history :

Revision Date	Version	Changes
2025-7	V2.0	Servo Stepper Version Manual Released
2025-12	V2.1.0	Cable section revised; cable drawings added; download links supplemented; Modbus protocol in Chapter 4 verified and updated.

Release Notes: Version V2 is a servo stepper communication protocol.

The following table lists the applicable Changingtek actuator models:

Actuator Operation Manual Servo Stepper	Compatible Models
Collaborative 2-finger Gripper	**CTAG2F90D**
Other models will be released successively.	

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Safety Precautions

Security Statement

警 告

- **Please strictly adhere to safety precautions. Incorrect operation may result in damage to this product and personal injury.**

- **High-risk groups: new employees (unskilled operators), maintenance personnel (who need to be in close contact with the gripper), and non-operators (who accidentally enter the work area) .**
- **During operation , it is strictly forbidden to put your hands, feet or other parts of your body into the clamping area of the equipment to avoid accidental pinching.**

Unpacking and inspection

警 告

- **If you find any damage, rust, or signs of use on the product or its accessories upon unpacking, please do not install it!**
- **Do not install the product if you find water inside, missing parts, or damaged parts when you open the box!**
- **Please carefully check the packing list. If you find any discrepancies between the packing list and the product name, please do not install!**

注 意

- **Before opening the box, please check whether the outer packaging of the equipment is intact and whether there is any damage, wetness, dampness, deformation, etc.!**
- **When unpacking, please check the surface of the equipment and accessories for any damage, rust, or dents!**
- **After unpacking, please carefully check the packing list to verify the quantity of equipment and accessories and whether the documents are complete!**

I. Cable Description

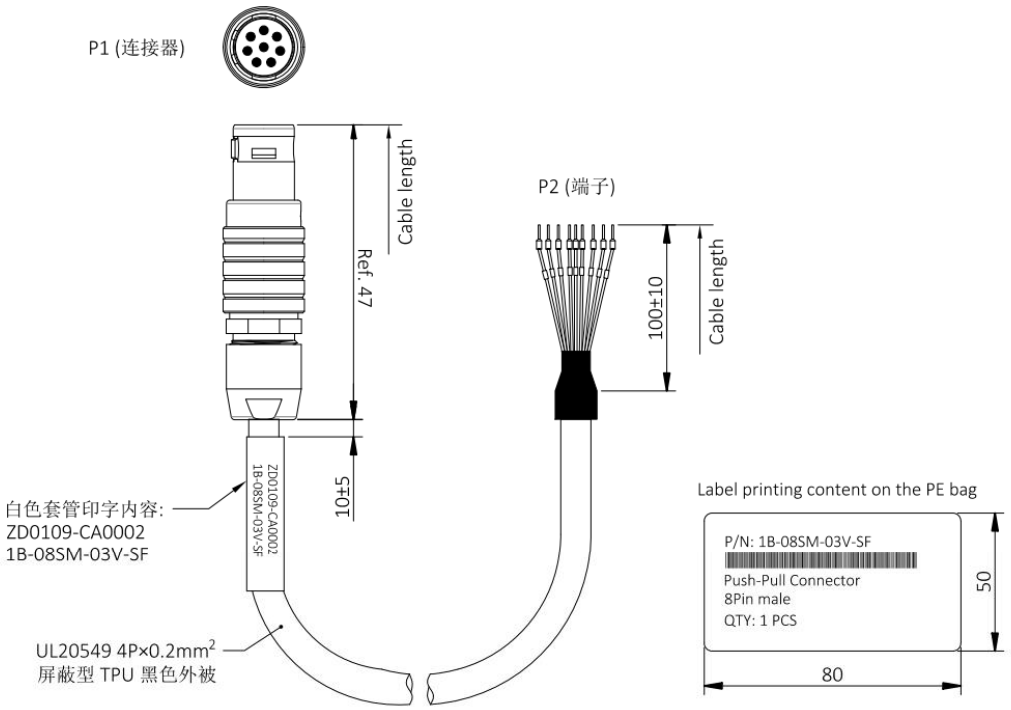
1.1 Cable Usage Instructions

The user connects the wires according to the wire marker definitions:

For IO mode control, refer to section 1.2.1 for wiring instructions; for 485 mode control, refer to section 1.2.2 for wiring instructions.

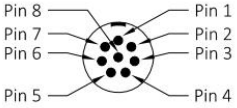
IO and 485 cannot be used simultaneously.

The standard 9-core cable (including shielding) diagram is as follows:

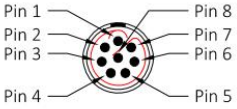


接线表

P1 (连接器)			字码管内容	端子型号
PIN 1	White	白 (0.2mm²)	IO_IN1	E0508管型端子
PIN 7	Brown	棕 (0.2mm²)	IO_IN2	同上
PIN 6	Green	绿 (0.2mm²)	DO_SS/B	同上
PIN 5	Blue	蓝 (0.2mm²)	DI_SS/A	同上
PIN 4	Orange	橙 (0.2mm²)	IO_OUT2	同上
PIN 3	Yellow	黄 (0.2mm²)	IO_OUT1	同上
PIN 2	Black	黑 (0.2mm²)	GND	同上
PIN 8	Red	红 (0.2mm²)	VCC_24V	同上
外壳	Braid	编织	PE	同上



Pin位分布(前视图)
BFGJ.1B.308.CPARD72



Pin位分布(焊接面)
BFGJ.1B.308.CPARD72

Wire Marker Identification	Definition	Remark
VCC_24V	24V+	Power supply for actuator
GND	GND	
PE	PE	
IO_OUT1	Output 1	
IO_OUT2	Output 2	
IO_IN1	Enter 1	
IO_IN2	Enter 2	
DI_SS/A	IO input common terminal (0-24V)/485-A	The common IO terminal and RS-485 cannot be used simultaneously.
DO_SS/B	IO output common terminal (0-24V) / 485-B	

1.2 Wiring Instructions for IO and RS485

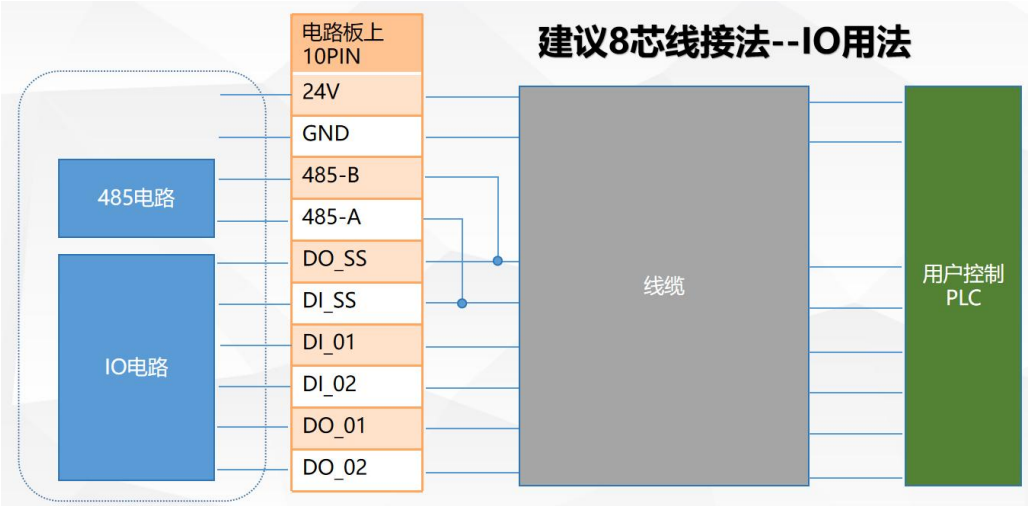
1.2.1 I/O Communication Wiring Instructions

The actuator wire markers DI_SS/A and DO_SS/B need to be connected to the input and output common terminals respectively. At the same time, the actuator IO_IN pin is connected to the output pin of the main control device's IO module.

The actuator IO_OUT pin corresponds to the input pin of the main control device's IO module.

Note: The actuator input and output support both NPN and PNP types.

The diagram is as follows:



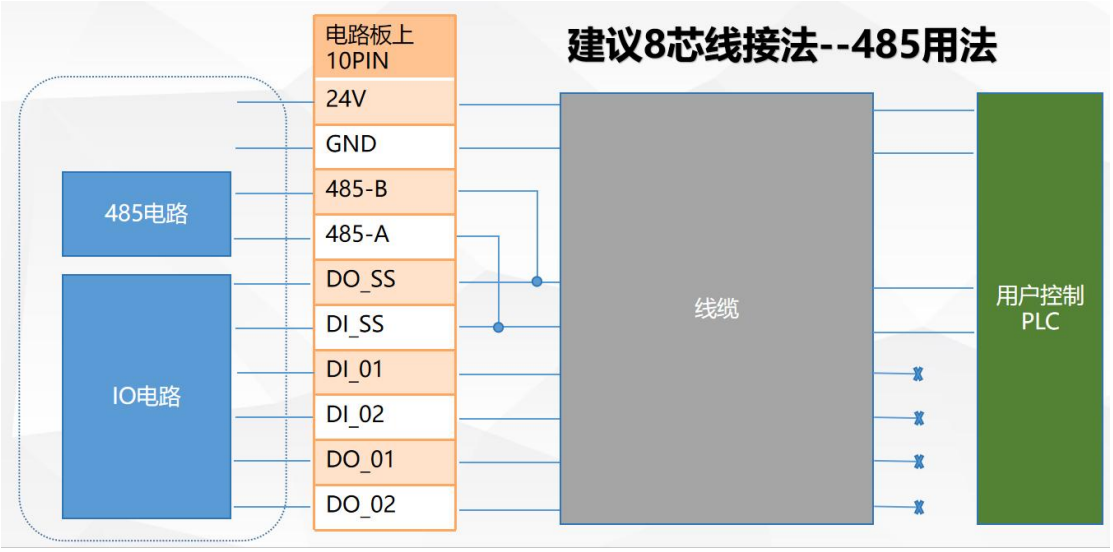
The IO wiring instructions are as follows:

Wire Marker Identification	Definition	Remark
VCC_24V	24V+	Power supply for actuator
GND	GND	
PE	PE	PE
IO_OUT1	Output 1	Connect to the main control device input terminal
IO_OUT2	Output 2	
IO_IN1	Enter 1	Connect to the output terminal of the main control device
IO_IN2	Enter 2	
DI_SS/A	IO input common terminal (0-24V)/485-A	Connect to common input terminal (Supports PNP and NPN)
DO_SS/B	IO output common terminal (0-24V) / 485-B	Connect to common output terminal (Supports PNP and NPN)

1.2.2 RS485 Communication Wiring Instructions

The actuator wire markers DI_SS/A and DO_SS/B need to be connected to the 485-A and 485-B ports of the main control device, respectively.

The diagram is as follows:



Wiring instructions are as follows:

Wire Marker Identification	Definition	Remark
VCC_24V	24V+	Power supply for actuator
GND	GND	
PE	PE	PE
IO_OUT1	Output 1	Not connected
IO_OUT2	Output 2	
IO_IN1	Enter 1	
IO_IN2	Enter 2	
DI_SS/A	IO input common	Connect to pin A of the host

	terminal (0-24V)/485-A	computer 485
DO_SS/B	IO output common terminal (0-24V) / 485-B	Connect to pin B of the host computer 485

II . I/O Communication Control

2.1 Introduction

The IO uses group input control. First, configure multiple point parameters in the host computer UServo Controller using 485 communication , and then switch to IO control mode to realize the control of the actuator by the master control device.

For example : the cable has IO_IN1 and IO_IN2, which supports controlling multiple points 1-4 , a total of 4 points;

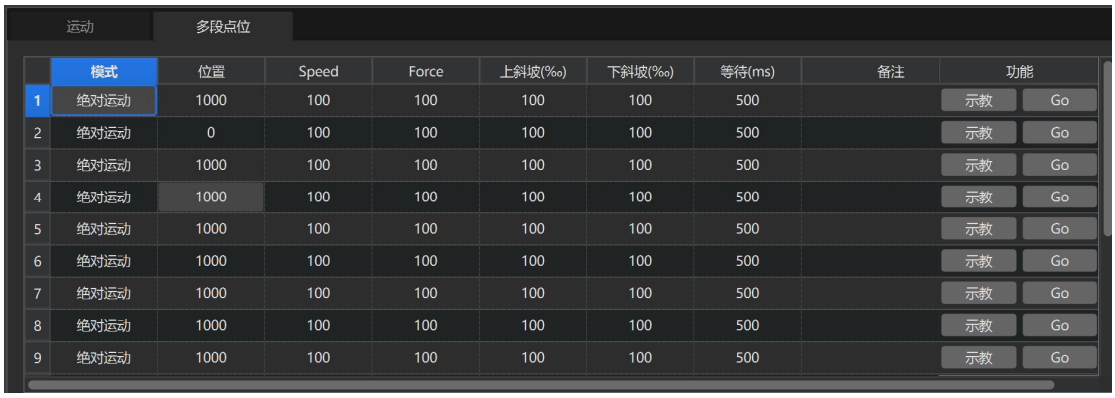
The control terminal connects to IO_IN1, and the host computer configures parameters for multi-segment point 1 and multi-segment point 2, which can complete the control of multi-segment point 1 and 2; the main control device triggers IO to complete the opening and closing action.

The group input definition table is as follows:

Serial Number	IO_IN 2	IO_IN 1	Definition	Remark
1	0	0	Multi-segment point 1	Corresponding to IO_IN 1
2	0	1	Multi-segment point 2	
3	1	0	Multi-	Connect

			segment point 3	to IO_IN 1 , IO_IN 2
4	1	1	Multi-segment point 4	

Note : The order of the **four points in the group input definition table corresponds one-to-one with the order of the first four multi-segment points in the host computer diagram below .**



2.2 Wiring

It is recommended to first connect to the 485 control mode, configure the multi-segment parameters of the UServo Controller host computer, and then switch to IO wiring to control the actuator.

For detailed instructions on I/O communication wiring, please refer to section **1.2.1 . IO communication wiring instructions** .

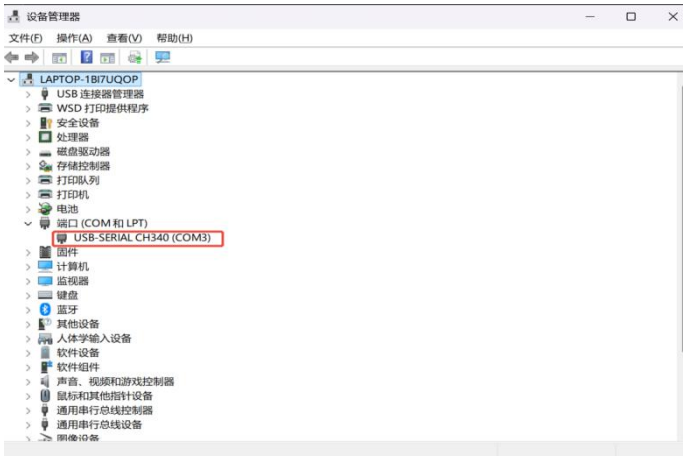
2.3 Multi-segment Configuration

UServo The controller's host computer is configured to perform multi-segment point operations as follows:

① Connect the actuator wiring harness number tubes DI_SS/A and DO_SS/B cables to the corresponding A+/D+ and B-/D- ports of the USB to 485 module. Insert the USB into the computer, open Device Manager, and check if the ports are displayed correctly.

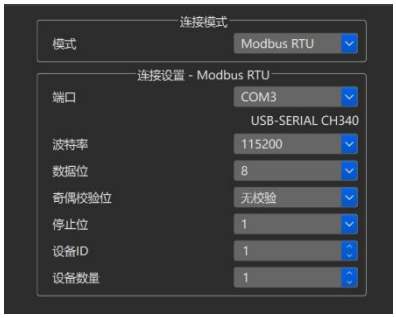
Note: For detailed instructions on 485 communication wiring, please refer to section

1.2.2, " 485 Communication Wiring Instructions".



A USB to 485 converter module is included with the shipment, as shown in the picture below:

② Install the host computer **UServo** In the **Controller** software, select the serial port number shown in the image above and establish a communication connection with the actuator. The connection settings are shown in the image below:



③ Open Control - Multi-segment Points (as shown in the figure below), and you need to configure the point parameters (position, speed, torque, uphill slope , downhill slope).

Location range: 0-1000‰ ;

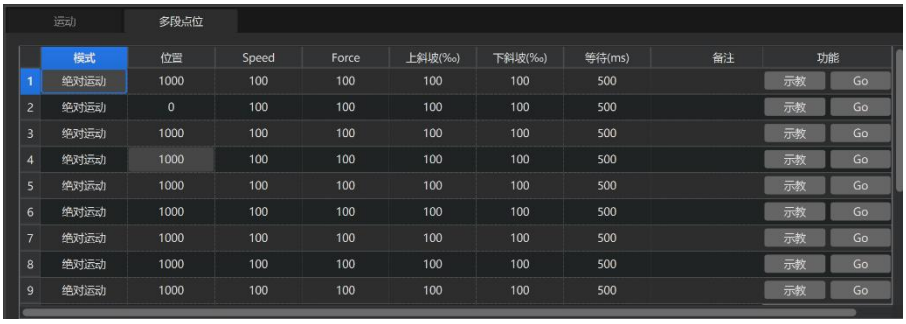
Speed range: 0-100%;

Torque range: 0-100%;

Uphill slope range: 0-100 0‰ ;

Downslope range: 0-100 0‰ ;

Note: It is recommended to set the clamping or external support torque to 60% and the opening torque to 100%.



运动 多段点位

	模式	位置	Speed	Force	上斜坡(‰)	下斜坡(‰)	等待(ms)	备注	功能	
1	绝对运动	1000	100	100	100	100	500		示教	Go
2	绝对运动	0	100	100	100	100	500		示教	Go
3	绝对运动	1000	100	100	100	100	500		示教	Go
4	绝对运动	1000	100	100	100	100	500		示教	Go
5	绝对运动	1000	100	100	100	100	500		示教	Go
6	绝对运动	1000	100	100	100	100	500		示教	Go
7	绝对运动	1000	100	100	100	100	500		示教	Go
8	绝对运动	1000	100	100	100	100	500		示教	Go
9	绝对运动	1000	100	100	100	100	500		示教	Go

2.4 I/O Control for Actuator Control

Completing the above IO physical wiring and teaching the multi- segment point operation to the host computer, the actuator is controlled by the main control device (PLC, robot, etc.).

① The actuator can be activated by triggering the corresponding OUT point output of the main control device.

② Observe the actuator feedback signal to see if the position or torque arrival signal is normal.

Notice:

The torque arrival signal will only be fed back when the torque reaches the set threshold.

The position arrival signal will only be fed back when the travel distance reaches the set position.

The position arrival signal can be affected by the size error of the workpiece being gripped; if the actuator movement is obstructed, such as by jamming or foreign objects, the torque will be reported as reached when the torque reaches the threshold.

III . Modus RTU Communication Control

3.1 Introduction

RS485 is a physical layer communication standard responsible for transmitting raw data between devices and the physical transmission medium. It handles the conversion of electrical signals to digital data.

It allows for the definition of voltage, timing, data rate, etc. It can support multiple devices communicating through the same bus, with one device acting as the master.

The instruction is received by other devices as slave devices.

Modbus RTU is an open serial communication protocol based on a master-slave architecture (i.e., client/server) originally developed by Schneider Electric. **The actuator uses the standard Modbus-RTU protocol and supports function codes 03, 06, and 10. Register address definitions are shown in Appendix 1.**

Function Code Name	Function Code	Broadcast Messages
Read holding register	0x03	Not supported
Write a single register	0x06	Supported
Write multiple registers	0x10	Supported

The actuator currently only supports the Modbus RTU protocol. If other protocols are required, a protocol conversion module can be purchased separately.

Modbus RTU message format:

Device ID	**Function Code**	**Register Address**	**Number of Registers**	**Byte Count**	**Data**	**Verification Code**

Device ID : Represents the actuator's ID number, which can be changed. The default actuator ID is 1.

Function code : Read and write operations on the executor register.

Register address : The corresponding register address for the actuator-related functions.

Number of registers : The number of registers written to or read from.

Number of bytes : Each register is 16 bits (2 bytes) in size, and N registers require 2N bytes.

Data : The specific data written to the registers, with each register consisting of 16 bits.

CRC checksum : Ensures that terminal devices do not accept data that has changed during transmission, thus ensuring system security and efficiency. CRC checksum uses...

The 16-bit cyclic redundancy check method involves conversion based on the preceding data. See http://www.ip33.com/crc.html for details. Perform the calculation.

3.1.1 Temporary Zone Control

The main control device can use the Modbus RTU protocol to control the actuators, and can adopt multi-segment point control and temporary area control.

Multi-segment point control : Retrieves the corresponding multi-segment point movement after being taught in the host computer, supporting up to 16 points, without the need to connect physical I/O lines.

Temporary zone control : The master control device sends target position, velocity, acceleration, deceleration, and target torque commands to the actuator, and then sends motion trigger commands.

This can trigger the actuator to move without the need to connect physical I/O lines.

Note: Modbus RTU recommends using temporary area control, and there is no limit to the number of temporary area points.

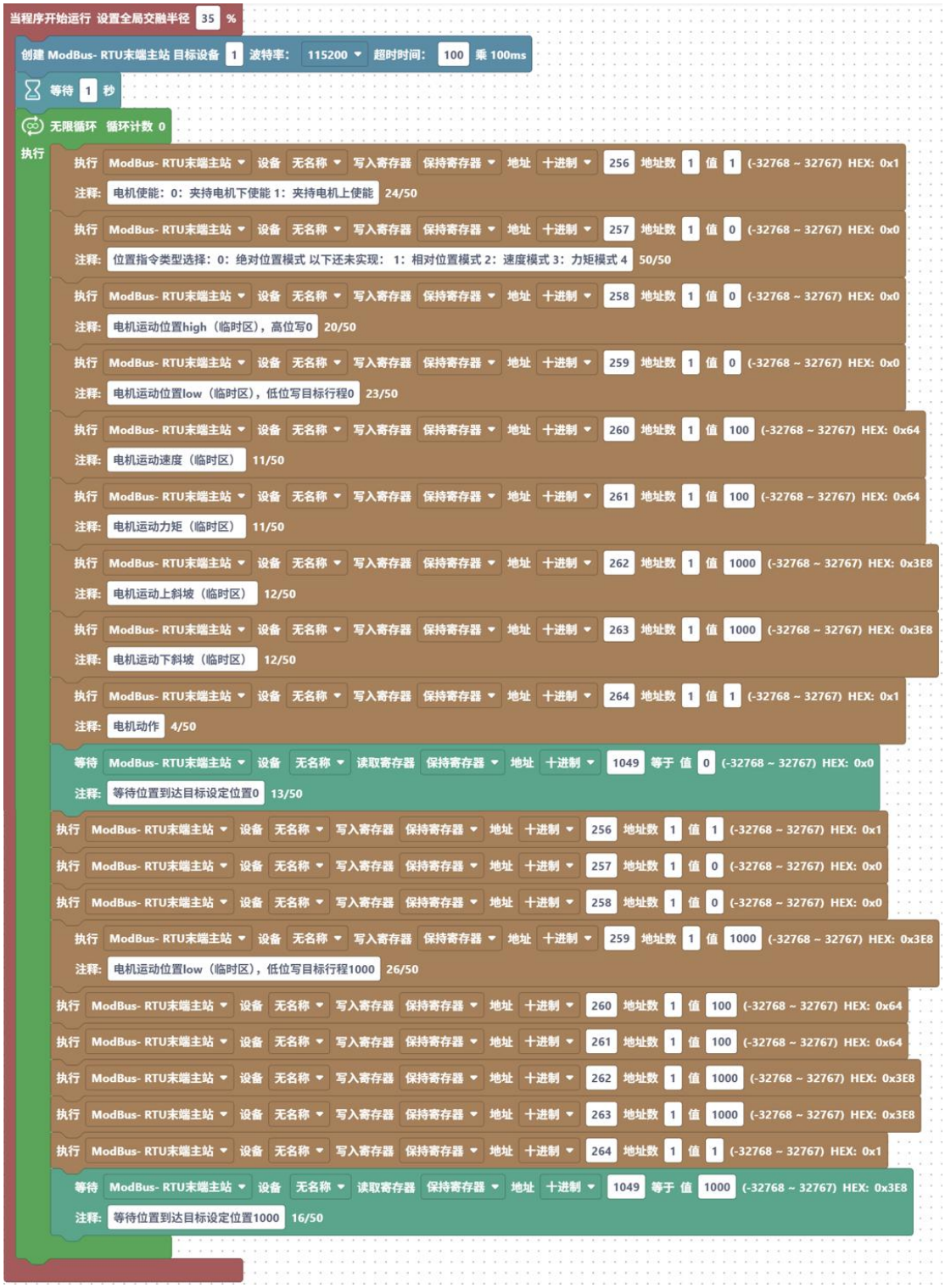
3.2 Wiring

For detailed instructions on 485 communication wiring, please refer to section **1.2.2 , "485 Communication Wiring Instructions".**

3.3 Control Example

3.3.1 Modbus RTU Temporary Area Control

Example as follows: (Taking CTAG2F90D and Realman robot control as an example, the maximum mapped position value is 1000)



IV . Appendix (Modbus Protocol)

1. Actuator Enable

This command is related to executor enable, and its address is 0x0100. This register address is readable and writable; when the register data is 0, the executor is disabled, and when it is 1, the executor is enabled.

Data sent: 01 06 01 00 00 01 49 F6

Returned data: 01 06 01 00 00 01 49 F6

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 00	00 01	49 F6

2. Actuator Movement Position (Temporary Zone)

This command specifies the temporary location of the executor, at addresses 0x0102 and 0x0103. The register address is a read-write address, and the data within the register represents the final location where the executor will run.

Data sent: 01 10 01 02 00 02 04 00 00 00 C8 7E 70

Returned data: 01 10 01 02 00 02 E1 F4

Device ID	Function Code	Register Address	Number Of Registers	Byte Count	Data	Verification Code
01	10	01 02	00 02	04	00 00 00 C8	7 E 70

This instruction sets the temporary motion position of the actuator to 200 (0x00 C8), (200 ‰) .

3. Actuator Movement Speed (Temporary Zone)

This command specifies the temporary movement speed of the executor, at address 0x0104. This register address is readable and writable, and the data within the register

is a percentage of the executor's maximum speed (0x0305).

Data sent: 01 06 01 04 00 64 C8 1C

Returned data: 01 06 01 04 00 64 C8 1C

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 04	00 64	C8 1C

This instruction sets the temporary motion speed of the actuator to 100 % (0x0064).

4. Actuator Movement Torque (Temporary Zone)

This command sets the temporary motion torque for the actuator at address 0x0105. This register address is readable and writable, and the data within the register is a percentage of the actuator's maximum torque (0x0306).

Data sent: 01 06 01 05 00 64 99 DC

Returned data: 01 06 01 05 00 64 99 DC

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 05	00 64	99 DC

This instruction sets the temporary motion torque of the actuator to 100 % (0x0064).

5. Actuator Movement Up Ramp (Temporary Zone)

This command is used for the temporary upward ramp motion of the actuator, with the address 0x0106. This register address is readable and writable, and the data within the register corresponds to the final upward ramp value for the actuator's operation.

Data sent: 01 06 01 06 0 3 E8 68 89

Returned data: 01 06 01 06 0 3 E8 68 89

Device ID	Function Code	Register Address	Data	Verification Code

01	06	01 06	03 E8	68 89

the temporary motion ramp of the actuator to 1000 ‰ (0x03E8).

6. Actuator Movement Down Ramp (Temporary Zone)

This command is used for the temporary downward ramp motion of the actuator, with the address 0x0107. This register address is readable and writable, and the data within the register corresponds to the final downward ramp value for the actuator's operation.

Data sent: 01 06 01 0 7 0 3 E8 39 49

Returned data: 01 06 01 0 7 0 3 E8 39 49

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 07	0 3 E8	39 49

the temporary motion deceleration of the actuator to 1,000 ‰ (0x03E8).

7. Temporary Zone Movement Triggered (0x0108)

This command is related to triggering movement in the temporary area, and its address is 0x0108. This register address is readable and writable. When the data in the register is 0, movement is not triggered. When it is 1, the executor runs with parameters from 0x0104 to 0x0107 to the movement position in the temporary area set at 0x0102-0x0103 .

Data sent: 01 06 01 08 00 01 C8 34

Returned data: 01 06 01 08 00 01 C8 34

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 08	00 01	C8 34

8. Alarm Reset (0x01 10)

This instruction is an alarm reset command, located at address 0x0110 . This register is a read-write address; when the data in the register is 0, the system performs no operation; when it is 1, the actuator resets the relevant alarms in 0x0402-0x0403.

Data sent: 01 06 01 10 00 01 48 33

Returned data: 01 06 01 10 00 01 48 33

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 10	00 01	48 33

9. Power-On Reset Switch - FLASH Area (0x01 13)

This instruction is a power-on reset instruction, located at address 0x0113 . This register is a read-write address; when the data in the register is 0, the executor will not automatically reset to zero upon power-on; when the data in the register is 1, the executor will automatically reset to zero upon power-on .

Data sent: 01 06 01 13 00 01 B8 33

Returned data: 01 06 01 13 00 01 B8 33

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 13	00 01	B8 33

10. Power-on Enable Switch - FLASH Area (0x01 14)

This instruction is a power-on enable instruction, with address 0x0114 . This register is a read-write address; when the data in the register is 0, the actuator will not be automatically enabled upon power-on; when the data in the register is 1, the actuator will be automatically enabled upon power-on .

Data sent: 01 06 01 14 00 01 09 F2

Returned data: 01 06 01 14 00 01 09 F2

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 14	00 01	09 F2

11. Save All Parameters (0x01 1A)

This instruction saves all parameters (FLASH area) at address 0x01 1A. This register is readable and writable; by default, modified parameters are not saved after a power outage. To retain the current parameter settings upon power-on, the current register must be manually activated.

Data sent: 01 06 01 1A 00 01 68 31

Returned data: 01 06 01 1A 00 01 68 31

Device ID	Function Code	Register Address	Data	Verification Code
01	06	01 1A	00 01	68 31

12 . Motor Motion Status Feedback (0x0 401)

This command provides feedback on the motor's motion status , and its address is 0x0 4 0 1. This register address is read-only and is defined as follows:

bit1 (0x0001): Position reached (under certain conditions, this can be used as evidence that the object was not successfully clamped)

bit2 (0x0002): Speed reached (speed reached maximum speed)

bit3 (0x0004): Zero speed reached (motor speed is 0 after movement begins)

bit4 (0x0008): Torque reached (under certain conditions, this can be used as evidence

of successful object clamping)

bit5 (0x0010): Ready to complete (Ready to complete = Torque reached | Position reached)

bit15 (0x8000): Motor enable state

Data sent: 01 03 0 4 0 1 00 01 D4 FA

Device ID	Function Code	Register Address	Data	Verification Code
01	03	0 4 01	00 01	D4 FA

Returned data: 01 03 02 00 01 79 84

Device ID	Function Code	Data Length	Data	Verification Code
01	0 3	0 2	00 1D	79 4D

Explanation: Convert 0x001D to 16-bit binary: 0000 0000 0001 1101

bit1 0x0001 1 position reached ☑

bit2 0x0002 0 Speed reached ✕

bit3 0x0004 1 Zero speed arrival ☑

bit4 0x0008 1 Torque reached ☑

bit5 0x0010 1 Ready to complete ☑

bit15 0x8000 0 Motor enable status ✕

13 . Alarm Message A (0x0 402)

This command is for alarm message feedback , and its address is 0x0 4 0 2. This register address is read-only and is defined as follows:

bit0 (0x01): Alarm indicating position exceeds minimum limit

bit1 (0x02): Alarm indicating position exceeds maximum value

bit2 (0x04): Curve overshoot alarm

bit3 (0x08): Overspeed alarm

bit4 (0x10): Overcurrent alarm

bit5 (0x20): Overvoltage alarm

bit6 (0x40): Undervoltage alarm

bit7 (0x80): Over-temperature alarm

bit8 (0x100): Low temperature alarm

bit9 (0x200): Hardware alarm

bit10 (0x400): Motor not calibrated alarm

bit11 (0x800): SPI communication fault alarm

Alarm Result = Alarm Information & Alarm Type

Data sent: 01 03 0 4 0 2 00 01 24 FA

Device ID	Function Code	Register Address	Data	Verification Code
01	03	0 4 02	00 01	24 FA

Returned data: 01 03 02 00 30 B8 50

Device ID	Function Code	Data Length	Data	Verification Code
01	0 3	0 2	00 30	B8 50

Explanation: Convert 0x0030 to 16-bit binary: 0 000 0000 0001 1000

B it 4 0x00081 Overcurrent Alarm ☑

B it 5 0x00101 Overvoltage alarm ☑

14 . Alarm Message B (0x0403)

This command is for actuator drop alarm , address 0x0403 . This register address is read-only; a value of 1000 in the register indicates that the actuator workpiece has fallen.

Data sent: 01 03 0 4 0 3 00 01 75 3A

Device ID	Function Code	Register Address	Data	Verification Code
01	03	0 4 03	00 01	75 3A

Returned data: 01 03 02 10 00 B5 84

Device ID	Function Code	Data Length	Data	Verification Code
01	0 3	0 2	10 00	B5 84

15. Real-time Feedback of Motor User Position (0x0418-0x0419)

This command is a real-time position information feedback instruction. The position information is the actuator travel percentage, and the addresses are 0x0418 and 0x0419 . This register address is a read-only address, and the data in the register represents the actuator's real-time position information (actuator position = (real-time feedback position information high << 16) + real-time feedback position information low).

Data sent: 01 03 0 4 18 00 02 45 3C

Device ID	Function Code	Register Address	Data	Verification Code
01	03	04 18	00 0 2	45 3C

Returned data: 01 03 04 00 00 23 28 E3 1D

Device ID	Function Code	Data Length	Data	Verification Code
01	03	04	00 00 23 28	E3 1D

The instruction is to set the feedback actuator position to 9000 (90mm).

16 . Real-time Feedback of Motor User Speed (0x041A)

This command is a real-time feedback instruction for engine speed information, expressed as a percentage of engine speed, and its address is 0x041A . This register address is read-only, and the data within the register represents the actuator's real-time engine speed information.

Data sent: 01 03 0 4 1A 00 01 A4 FD

Device ID	Function Code	Register Address	Data	Verification Code
01	03	0 4 1A	00 01	A4 FD

Returned data: 01 03 02 00 50 B8 78

Device ID	Function Code	Data Length	Data	Verification Code
01	0 3	0 2	00 50	B8 78

This instruction sets the feedback motor speed to 80%.

17. Real-time Feedback of Motor User Current (0x041B)

This command is a real-time current feedback instruction, expressed as a percentage of the maximum current value, and its address is 0x041B . This register address is read-only, and the data within the register represents the actuator's real-time current information.

Data sent: 01 03 0 4 1B 00 01 F5 3D

Device ID	Function Code	Register Address	Data	Verification Code

01	03	0 4 1B	00 01	F5 3D

Returned data: 01 03 02 00 50 B8 78

Device ID	Function Code	Data Length	Data	Verification Code
01	0 3	0 2	00 50	B8 78

This instruction sets the current of the feedback motor to 80%.

V. Servo Stepper Data Download

The SDK download link is as follows:

http://58.210.198.4:5556/d/s/15z0hhCGaM5ewT1Rq2eUv0zCCWiDcQPd/OeFaoFWE2zewssiA_qW4KDQRZMJQROMc-a7tgoPXFxQw

The ROS download link is as follows:

https://gitee.com/Ice_Panda/changingtek

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